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Inventory & Stock Management System

Project Team

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Inventory & Stock Management System

1. Project Overview

This project is about creating a computer-based system to manage and track stock in an organization. The system will let users add new products, update details, and remove items from stock. It will also keep a record of how much stock is available in the organization, give alerts when stock is low in the organization, and store supplier information. Because while doing manual work people do a lot of mistakes. This system will help to reduce mistake and make inventory handling more faster and more accurate.

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2. Motivation

Many local and small businesses still use manual methods like notebooks or spreadsheets to manage inventory. This often causes problems for them like having too much stock, running out of items, or losing track of products. A proper system like this will help keep records safe, updated, and easy to access for these businesses. The main reason for making this project is to help businesses to save time, avoid mistakes, and make stock management simpler and more reliable.

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3. Objectives

The goals of this project are:

- To design a desktop application using JavaFX.
- To store and manage inventory data in a database using SQL.
- To add, update, or delete product details easily.
- To show alerts when stock is low.
- To keep supplier details linked with products.
- To generate reports like monthly stock use and purchase history.

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4. Expected Outcomes

After finishing this project, we expect to have:

- A simple and easy-to-use desktop system for managing inventory.

- Secure login for admin and staff.
- Real-time stock updates with alerts for low stock.
- Automatic reports for stock, suppliers, and product movement.
- A system that reduces manual work and improves accuracy.

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5. Project Scope

The system will include these features:

- **Product Management**: Add, update, delete, and search products.
- **Stock Management**: Record stock coming in and going out.
- **Supplier Management**: Store supplier details.
- **Alerts**: Show messages when stock is low.
- **Reports**: Create stock summary and supplier reports.

The system will not include online access, barcode scanning, or multiple branches. It will focus only on a single organization's stock management.

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6. Tools and Technologies

- **Frontend**: JavaFX for the application interface.
- **Backend Database**: SQL.
- **Development Environment**: IntelliJ or Eclipse.